Lanthanothermic Reduction of TiO_2

TAKARA TANAKA, TAKANARI OUCHI, and TORU H. OKABE

The development of a new titanium (Ti) smelting process is highly desired to decrease the cost of Ti metal. In this study, a new method to produce a low-oxygen-concentration Ti directly from TiO_2 using La as the reductant was developed. First, deoxidation experiments of metallic Ti using La as the deoxidant were conducted, and the ability of La to deoxidize metallic Ti was demonstrated. The oxygen concentrations in Ti were approximately 3900 ± 400 mass ppm O and 120 ± 50 mass ppm O at La/La_2O_3 and La/LaOCl/$LaCl_3$ equilibria, respectively, at 1200 K (927 °C). Then, TiO_2 reduction experiments utilizing La as the reductant were conducted to produce low-oxygen-concentration Ti directly. TiO_2 pellets and Ti pieces were held in $LaCl_3$ molten salt with a sufficient amount of La at 1200 K (927 °C) for 86 ks. After the reaction, the TiO_2 pellets were reduced to metallic Ti without reaction inhibition by the formation of intermediate products. The oxygen concentration of the Ti pieces, which were placed in the molten salt with the TiO_2 pellets, was 100 ± 50 mass ppm O. These results indicate that low-oxygen-concentration Ti containing less than 200 mass ppm O could be produced directly from TiO_2 by utilizing La as the reductant in $LaCl_3$ molten salt at 1200 K (927 °C). This new method is expected to lead to the development of a new industrial process for the production of ultra-low-oxygen-content Ti from Ti ore without a chlorination process.

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I. INTRODUCTION

THE high cost of the titanium (Ti) smelting process is one of the major factors of the high cost of Ti. Currently, industrial Ti smelting is mainly performed with the Kroll process (Figure 1), which is a multistage process involving the chlorination of Ti ore containing a large amount of Ti oxide to produce pure $TiCl_4$ and the reduction of the purified $TiCl_4$ using magnesium (Mg). It is a batch process that requires a long time for the reduction of $TiCl_4$ and the separation of Mg and $MgCl_2$ from Ti. Therefore, the process has low productivity and high costs.[1,2] Accordingly, the development of a new smelting process is highly desired to decrease the cost of Ti. To this end, the direct reduction of TiO_2 has been investigated extensively. Such a direct reduction process is expected to simplify and speed up the Ti smelting process; this is because Ti oxide, which is the main component of Ti ore, can be treated directly.

Because of the strong affinity of Ti with O, reductants that can directly reduce TiO_2 to metallic Ti with the industrially acceptable oxygen concentration of approximately 500 mass ppm O are thermodynamically limited to calcium (Ca) and some rare earth metals (REs).[3–5] In the past, many reduction processes using Ca as the reductant have been investigated, including the calciothermic reduction process,[6–16] and combinations of calciothermic and electrochemical reduction methods, such as the FFC[17–20] and OS processes.[21–24] However, none of these methods is currently applied in large-scale industry settings because of many technical problems, such as the inefficient regeneration of Ca and contamination with impurities, such as carbon. A number of studies have been conducted on the TiO_2 reduction using some metal reductants, such as Mg and Al, which are more efficiently produced and recycled than Ca.[25–30] However, metal (or alloy) reductants have an insufficient reducing ability and cannot stably produce Ti with oxygen concentrations of approximately 500 mass ppm O.

However, it is known that some REs can thermodynamically deoxidize metallic Ti to low-oxygen-concentration Ti.[31] Muthmann *et al.*[32,33] produced some refractory metals, such as tantalum (Ta) and tungsten (W), using a misch metal (*i.e.,* an alloy of several REs, including lanthanum (La), cerium (Ce), praseodymium (Pr), and neodymium (Nd)) as the reductant. Based on this method, Vogel *et al.*[34] tried to produce metallic Ti by direct reaction between a misch metal and TiO_2.

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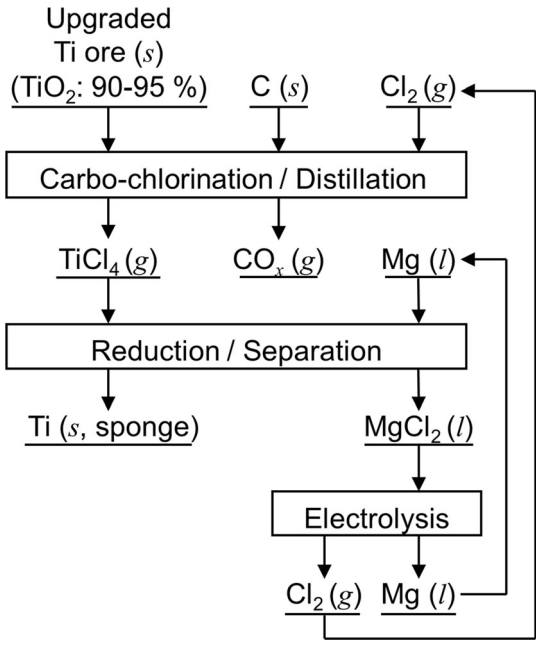


Fig. 1—The process flow of the Kroll process.

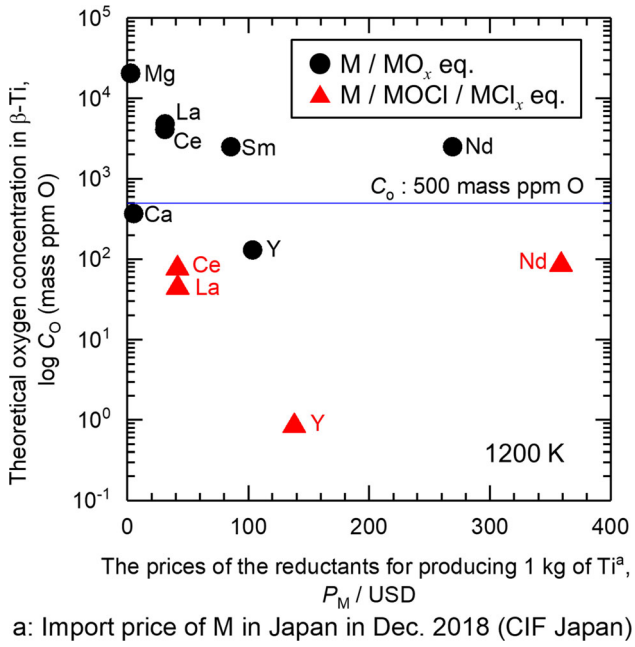


Fig. 2—The relationship between the prices of reductants, M, for producing 1 kg of Ti and the theoretical oxygen concentration in β-Ti in M/MO_x eq. or $M/MOCl/MCl_x$ eq. at 1200 K (927 °C).[5,41–46]

However, they reported that, although metallic Ti was obtained, the obtained metal contained large amounts of impurities, and a large amount of Ti nitride was produced. Eventually, they disregarded this method because of the poor Ti yield.

Until recently, the use of REs as reductants for TiO_2 had been considered economically non-feasible. However, owing to the rapid expansion of the rare earth alloy magnet industry and the establishment of stable smelting methods for REs, such as Nd, several REs are in oversupply. La and Ce, which are produced as by-products of Nd, have not found many applications and are in overproduction.[5,35] Therefore, the application of REs, such as La and Ce, is an important issue.

Recently, Okabe *et al.* showed that metallic Ti can be thermodynamically deoxidized to low-oxygen-concentration Ti by REs.[5,36] Some REs, such as La and Ce, have relatively poor deoxidation abilities at metal/metal oxide equilibria compared to Ca; however, they can obtain superior deoxidation abilities to Ca by utilizing the formation of their oxyhalides. More recently, the effect of the oxychloride formation on the deoxidation ability for Ti was verified experimentally for yttrium (Y) and holmium (Ho).[37–40]

Figure 2 shows the relationship between the theoretical deoxidation ability of reductant M for metallic Ti at 1200 K (927 °C) and the cost of M for producing 1 kg of Ti from TiO_2.[41,42] The theoretical deoxidation ability was calculated using previously reported thermodynamic data.[5,43–46] Figure 2 indicates that La and Ce are excellent reductants in terms of both deoxidation ability and cost.

Based on the above consideration, La was employed as the reductant in this study. La has already been used in industry as one of the reductants for oxide reduction. Currently, La is used as a reductant in the metallothermic reduction of some rare earth oxides (lanthanothermic reduction). The combination of this method with vacuum distillation is referred to as the reduction–distillation method; during this process, the oxides of some REs, including samarium (Sm), europium (Eu), thulium (Tm), and ytterbium (Yb), are reduced by La, and high-purity REs are obtained by exploiting the difference in vapor pressure between the REs and La.[47–50]

In this study, a new method to produce low-oxygen-concentration Ti directly from TiO_2 using La as the reductant was developed. The ability of La to deoxidize metallic Ti was evaluated experimentally, and a lanthanothermic TiO_2 reduction experiment was conducted.

II. THERMODYNAMIC CALCULATION

A. *Deoxidation Ability of La for Metallic Ti*[5]

Ti has an extremely high affinity with O, and O has a high solubility in metallic Ti. Therefore, to produce Ti containing industrially suitable oxygen concentrations (500 mass ppm O) directly from TiO_2, the reductant is required to have a high potential to deoxidize metallic Ti.

Recently, based on a thermodynamic assessment, Okabe *et al.* proposed that metallic Ti can be deoxidized effectively by the formation of rare earth oxychlorides (REOCls), such as LaOCl and YOCl.[5] The effect of Y as the deoxidant was verified experimentally, and Ti with an extremely low oxygen concentration of approximately 30–60 mass ppm O was obtained.[51] On the other hand, the deoxidation ability of La was not experimentally verified.

Figure 3 shows a p_{O_2}-p_{Cl_2} phase stability diagram of a LaOCl system at 1200 K (927 °C).[5] Table I shows the values of ΔG_f° of each chemical species used for drawing Figure 3. At La/La_2O_3 and La/LaOCl/$LaCl_3$ equilibria, the oxygen concentration in Ti is determined by Eqs. [1] and [2] and Eqs. [3] and [4], respectively.

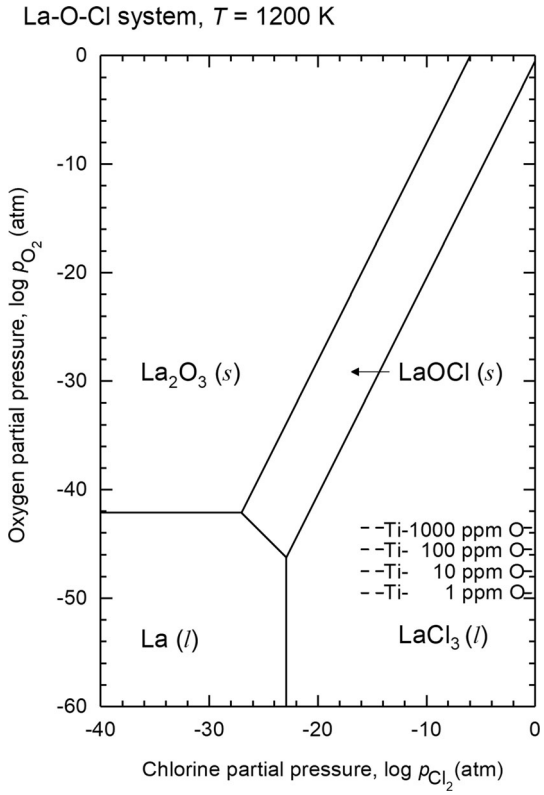


Fig. 3—p_{O_2}-p_{Cl_2} phase stability diagram of LaOCl systems at 1200 K (927 °C).

$$\mathrm{O\,(in\,Ti)} + \frac{2}{3}\,\mathrm{La}(l) = \frac{1}{3}\,\mathrm{La_2O_3}(s), \qquad [1]$$

$$\begin{aligned}\Delta G^\circ_{r,1} &= \frac{1}{3}\Delta G^\circ_{f,La_2O_3} - \Delta G^\circ_{1,Ti} \\ &= -2.303\,RT\log\frac{{a_{La_2O_3}}^{1/3}}{f_O\cdot[O]_{Ti}\cdot {a_{La}}^{2/3}},\end{aligned} \qquad [2]$$

$$\mathrm{O\,(in\,Ti)} + \frac{2}{3}\,\mathrm{La}(l) + \frac{1}{3}\,\mathrm{LaCl_3}(l) = \mathrm{LaOCl}\,(s), \qquad [3]$$

$$\begin{aligned}\Delta G^\circ_{r,3} &= \Delta G^\circ_{f,LaOCl} - \frac{1}{3}\Delta G^\circ_{f,LaCl_3} - \Delta G^\circ_{1,Ti} \\ &= -2.303\,RT\log\frac{a_{LaOCl}}{f_O\cdot[O]_{Ti}\cdot {a_{La}}^{2/3}\cdot {a_{LaCl_3}}^{1/3}},\end{aligned} \qquad [4]$$

where ΔG°_r is the standard Gibbs energy change of the reaction, $\Delta G^\circ_{f,i}$ is the standard Gibbs energy of the formation of compounds i, and a_i is the activity of chemical species i. $\Delta G^\circ_{1,Ti}$ is the standard Gibbs energy of oxygen dissolution in β-Ti, as shown by Eqs. [5] and [6].[46]

$$\frac{1}{2}\,\mathrm{O_2}(g) = \mathrm{O\,(in\,Ti)} \qquad [5]$$

$$\begin{aligned}\Delta G^\circ_{1,Ti} &= -2.303RT\log\frac{f_O\cdot[O]_{Ti}}{{p_{O_2}}^{1/2}} \\ &= -583,000 + 88.5T\ [1173-1373\ \mathrm{K}]\end{aligned} \qquad [6]$$

Table I. Standard Gibbs Energy of Formation of Some Compounds, and the Standard Gibbs Energy of Oxygen Dissolution in β-Ti

Compound, i	Standard Gibbs Energy of Formation, $\Delta G^\circ_{f,i}$/J mol^{-1}		$\Delta G^\circ_{f,i}$ at 1200 K/J mol^{-1}	Refs.
La_2O_3 (s)	$\Delta G^\circ_{f,i} = -1{,}794{,}000 + T\times 284.8$	1100 to 1500 K (827 to 1227 °C)	− 1,452,000	[43]
$LaCl_3$ (l)	$\Delta G^\circ_{f,i} = -989{,}400 + T\times 167.0$	1131 to 1500 K (858 to 1227 °C)	− 789,000	[43]
LaOCl (s)	$\Delta G^\circ_{f,i} = -1{,}002{,}000 + T\times 172.8$	1000 to 1200 K (727 to 927 °C)	− 794,600	[43]
TiO_2 rutile (s)	$\Delta G^\circ_{f,i} = -939{,}800 + T\times 177.2$	1173 to 1500 K (900 to 1227 °C)	− 727,200	[43]
Ti_4O_7 (s)	$\Delta G^\circ_{f,i} = -3{,}373{,}000 + T\times 595.6$	1173 to 1500 K (900 to 1227 °C)	− 2,658,000	[43]
Ti_3O_5 (s)	$\Delta G^\circ_{f,i} = -2{,}428{,}000 + T\times 417.2$	1173 to 1500 K (900 to 1227 °C)	− 1,927,000	[43]
Ti_2O_3 (s)	$\Delta G^\circ_{f,i} = -1{,}496{,}000 + T\times 255.7$	1173 to 1500 K (900 to 1227 °C)	− 1,189,000	[43]
TiO (s)	$\Delta G^\circ_{f,i} = -532{,}900 + T\times 87.8$	1265 to 1500 K (900 to 1227 °C)	− 427,600	[43]
[O] in Ti[a]	$\Delta G^\circ_{f,i} = -583{,}000 + T\times 88.5$	1173 to 1373 K (900 to 1100 °C)	− 476,800	[46]

[a]1/2 O_2 (g) = [O] in Ti (1 mass pct), Henrian 1 mass pct standard.

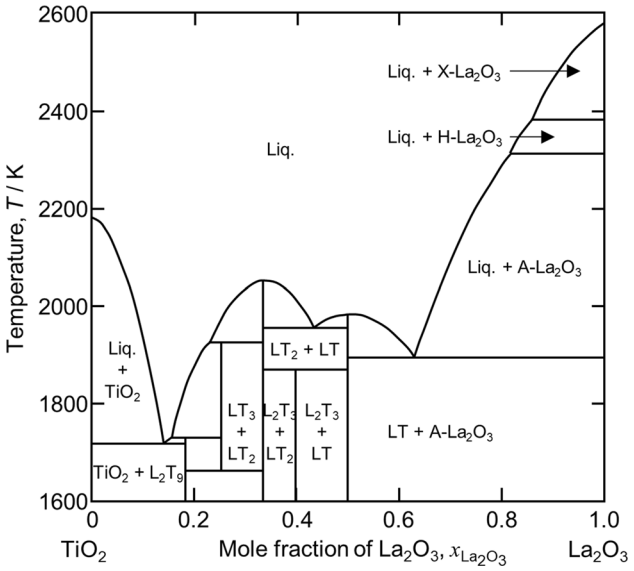


Fig. 4—TiO_2-La_2O_3 pseudo-binary phase diagram.[54].

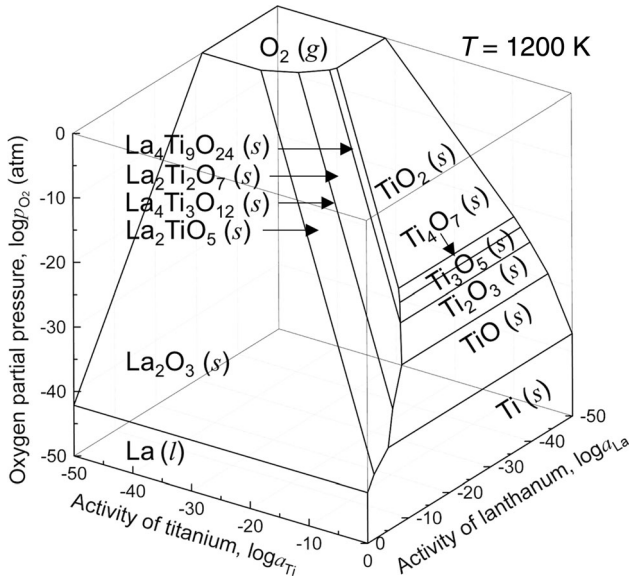


Fig. 5—Chemical potential diagram of La-Ti-O ternary system at 1200 K (927 °C).

where $[O]_{Ti}$ (mass pct O) is the equilibrium concentration of oxygen in Ti and f_O is the Henrian activity coefficient of the oxygen dissolved in β-Ti. The oxygen dissolution in Ti follows Henry's law and the value of f_O is considered to be unity. (The standard state of oxygen in Ti is the Henrian 1 mass pct O standard.)

The oxygen concentrations in Ti at La/La_2O_3 and $La/LaOCl/LaCl_3$ equilibria at 1200 K (927 °C) are calculated to be 4800 mass ppm O and 40 mass ppm O, respectively, from Eqs. [2], [4], and [6].

However, the thermodynamic data of rare earth compounds are generally unreliable. In addition, the stability of LaOCl in high-temperature molten salt has not been confirmed. It was reported that LaOCl decomposes in vacuum at 1207 K (934 °C)[52] while it is stable in Ar atmosphere at 1223 K (950 °C) for 4 h. Furthermore, thermodynamic data of LaOCl have only been reported for temperatures of up to 1200 K (927 °C)[53] Therefore, it is necessary to conduct a deoxidation experiment of metallic Ti using La as the reductant to verify the result of the thermodynamic calculation.

B. *Reduction Ability of La for TiO_2*

In general, the reduction reaction from oxide to metal may be prevented by intermediate products generated during the reduction. Particularly, when TiO_2 is reduced by La, complex oxides of the La-Ti-O system can form; many types of complex oxides of the La-Ti-O system have been reported. The La_2O_3-TiO_2 pseudo-binary system phase diagram is shown in Figure 4, and the thermodynamic data of the complex oxides have also been reported.[54]

A chemical potential diagram of the La-Ti-O ternary system at 1200 K (927 °C) is shown in Figure 5. Tables I and II show the ΔG_f° and ΔG_r° used to draw Figure 5, respectively. Only the complex oxides of the La_2O_3-TiO_2 system, of which thermodynamic data were available, were considered as the complex oxides of the La-Ti-O ternary system. As shown in Figure 5, the $La/La_2O_3/Ti$ three-phase equilibrium exists at 1200 K. The figure indicates that, when a sufficient amount of metallic La is present in the system (*i.e.,* when the activity of La is unity), the reduction of TiO_2 to Ti is feasible according to the results obtained by thermodynamic calculations (Eq. [7]).

$$TiO_2(s) + \frac{4}{3}\,La\,(l) = Ti\,(s) + \frac{2}{3}\,La_2O_3(s) \qquad [7]$$

In addition, based on the thermodynamic consideration made in Section II–A, it can be deduced that, when adding $LaCl_3$ molten salt in the system, TiO_2 is not only reduced to metallic Ti but can also be reduced directly to low-oxygen-concentration Ti by utilizing the LaOCl formation reaction.

III. EXPERIMENTAL PROCEDURE

Figure 6 shows a schematic of the experimental apparatus. For the deoxidation experiment of metallic Ti, La shots (≥ 99 pct, Shot, Santoku Co., Ltd.), La_2O_3 powder (≥ 99.9 pct, Kojundo Chemical Laboratory Co., Ltd.), $LaCl_3$ powder (≥ 99.9 pct (REO), ultra dry, Alfa Aesar), and Ti pieces for measuring the oxygen potential in the system were placed in Ti crucibles (CP-Ti, 6.35 mm diameter × 0.7 mm thickness × 80 mm length, Toho Technical Service Co., Ltd.) as shown in Figures 6(a) and (b). The tops of the Ti crucibles were sealed by welds. Three types of Ti pieces with different shapes and oxygen concentrations (Ti-1.2U: CP-Ti wire, 1.2 mm diameter × 20 mm length, 1300 mass ppm O, Shinkinzoku Industry Co., Ltd.; Ti-2.0I: CP-Ti wire, 2.0 mm diameter × 10 mm length, 900 mass ppm O, Shinkinzoku Industry Co., Ltd.; Ti-2B: Ti cube, 2 mm in length of side, 230 mass ppm O, electron-beam-melted high-purity Ti sponge) were used.

Table II. Standard Gibbs Energy of Reaction of Some Compounds in La-Ti-O System

Compound, i	Reaction Equation Standard Gibbs Energy of Reaction, $\Delta G^\circ_{r,i}$/J mol^{-1}		$\Delta G^\circ_{r,i}$ at 1200 K/J mol^{-1}	Refs.
$La_4Ti_9O_{24}$ (s)	2 La_2O_3 (s) + 9 TiO_2 (s) = $La_4Ti_9O_{24}$ (s) $\Delta G^\circ_{r,i} = -603{,}900 - T \times 0.28$	1173 to 1500 K (900 to 1227 °C)	− 604,200	[54]
$La_2Ti_2O_7$ (s)	La_2O_3 (s) + 2 TiO_2 (s) = $La_2Ti_2O_7$ (s) $\Delta G^\circ_{r,i} = -248900 - T \times 21.27$	1173 to 1500 K (900 to 1227 °C)	− 274,400	[54]
$La_4Ti_3O_{12}$ (s)	2 La_2O_3 (s) + 3 TiO_2 (s) = $La_4Ti_3O_{12}$ (s) $\Delta G^\circ_{r,i} = -447{,}000 - T \times 15.91$	1173 to 1500 K (900 to 1227 °C)	− 466,100	[54]
La_2TiO_5 (s)	La_2O_3 (s) + TiO_2 (s) = La_2TiO_5 (s) $\Delta G^\circ_{r,i} = -198{,}100 + T \times 5.36$	1173 to 1500 K (900 to 1227 °C)	− 191,700	[54]
$La_2Ti_3O_9$ (s)	La_2O_3 (s) + 3 TiO_2 (s) = $La_2Ti_3O_9$ (s) $\Delta G^\circ_{r,i} = -126{,}600 - T \times 99.07$	1173 to 1500 K (900 to 1227 °C)	− 245,500	[54]

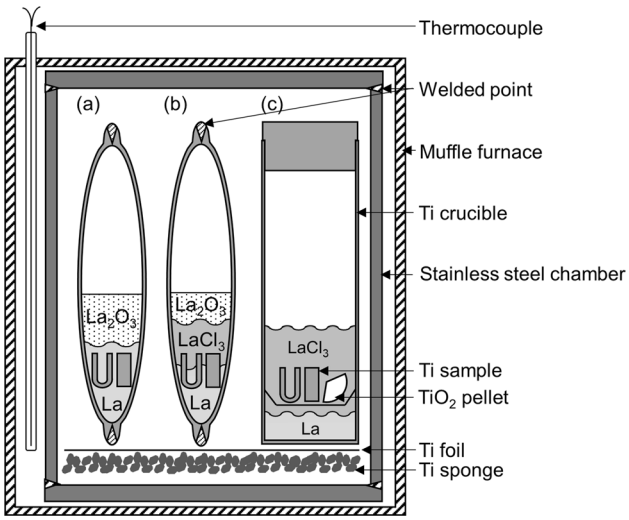


Fig. 6—Schematic of the experimental apparatus (T = 1200 K, t' = 86 to 173 ks); (*a*) deoxidation experiment of metallic Ti in La/La_2O_3 eq., (*b*) deoxidation experiment of metallic Ti in La/LaOCl/$LaCl_3$ eq., and (*c*) reduction experiment of TiO_2 in La/LaOCl/$LaCl_3$ eq..

For the TiO_2 reduction experiment, TiO_2 powder (≥ 98 pct, Anatase, Kanto Chemical Co., Ltd.) was sintered into pellets so that the reduction state of TiO_2 could be easily observed. TiO_2 powder was pressed at approximately 600 MPa, heated at 1300 K (1027 °C) for 86 ks in air, and sintered. The TiO_2 pellets, La shot, $LaCl_3$ powder, and Ti pieces were placed in a Ti crucible (CP-Ti, 25.4 mm diameter × 1 mm thickness × 80 mm length, Shinkinzoku Industry Co., Ltd.) as shown in Figure 6(c). As shown in Figure 6, La was placed at the bottom of the Ti crucible, and Ti foil was placed on it. The TiO_2 pellets and Ti pieces were put on the Ti foil to avoid direct contact with La.

The Ti crucibles together with a small amount of Ti sponge (≥ 99 pct, Toho Titanium Co., Ltd.) were placed in a stainless-steel crucible (SUS-316, 88.9 mm diameter × 3 mm thickness × 100 mm length, Azabu Industry Co., Ltd.), which was completely sealed by welds. Then, the stainless-steel crucible was left in a muffle furnace at 1200 K (927 °C) and 1300 K (1027 °C) for 86–173 ks and was quenched in water. The stainless-steel and Ti crucibles were opened mechanically, and samples were taken out.

The Ti pieces were cleaned by leaching first with distilled water and then with diluted HCl solution (1 + 10) (HCl: 35.0–37.0 pct, Kanto Chemical Co., Ltd.). Ti pieces were first chemically polished with a 1:4:10 mixture of HF-HNO_3-H_2O (HF: 46.0–48.0 pct, Morita Chemical Industry Co., Ltd.; HNO_3: 60.0–62.0 pct, Wako Industry Co., Ltd.) and then rinsed with distilled water and acetone (≥ 99 pct, Kanto Chemical Co., Ltd.). The oxygen concentrations of the Ti pieces, which had different shapes and initial oxygen concentrations, were analyzed using the inert gas fusion technique (TC-600, LECO Corporation). When their oxygen concentrations were approximately the same, the system was considered to have reached equilibrium.

The pellets taken out were also cleaned by leaching with distilled water and diluted HCl solution. Finally, an X-ray diffraction (XRD) analysis (D2 Phaser, Bruker) of the removed salt and the cleaned pellets was conducted.

IV. RESULTS AND DISCUSSION

A. *Deoxidation Experiment of Metallic Ti*

Table III shows the conditions under which each sample was deoxidized at La/La_2O_3 or La/LaOCl/$LaCl_3$ equilibrium and the resulting oxygen concentrations of the Ti pieces. Figure 7 shows the cross-section of the Ti crucible after heating. It can be seen that the Ti pieces were immersed in the La metal and the molten salt.

Figure 8 shows the relationship between the deoxidation results of the Ti pieces and the molar fraction of La_2O_3, with $x_{La_2O_3}$ at La/La_2O_3 equilibrium at 1200 K (927 °C); $x_{La_2O_3}$ is defined as $x_{La_2O_3} = \frac{n_{La_2O_3}}{n_{La_2O_3} + n_{La}}$, where n_i (mol) is the initial amount of chemical species i. The oxygen concentration in Ti increased to approximately 3900 ± 400 mass ppm O, as expected from the thermodynamic calculation (Table III and Figure 8). Furthermore, as can be seen in Figure 8, there was no increase or decrease in the oxygen concentration of Ti

Table III. Results of Oxygen and Nitrogen Analyses of Ti Samples in the Experiment Utlizing La/La_2O_3 eq. and $La/LaOCl/LaCl_3$ eq.

Exp. No.	Initial Quantity of Samples in the Titanium Crucible*, m/g (n/mol)			Experimental Condition**	Type of Ti Samples†	Oxygen Concentration, C_O (mass ppm O)		Analyzed Error, ε_O (pct)	Nitrogen Concentration, C_N (mass ppm N)		Analyzed Error, ε_N (pct)
	La	La_2O_3	$LaCl_3$			Initial, $C_{O,\ initial}$	After Exp., $C_{O,\ after}$		Initial, $C_{N,\ initial}$	After exp., $C_{N,\ after}$	
N4¶	4.99	1.00		1200 K/173 ks	Ti-1.2U	1300	3600	1.9	30	50	11
	(0.036)	(0.0031)		$x_{La_2O_3} = 0.079$	Ti-2.0I	900	4200	1.8	40	80	9.7
N6¶	5.01	4.00		1200 K/173 ks	Ti-1.2U	1300	3700	1.9	30	70	9.9
	(0.036)	(0.012)		$x_{La_2O_3} = 0.25$	Ti-2.0I	900	4000	1.8	40	100	9.5
F3§	11.5	2.50		1300 K/173 ks	Ti-1.2U	1300	3100	2.4	30	350	9.5
	(0.083)	(0.0077)		$x_{La_2O_3} = 0.085$	Ti-2.0I	900	3100	2.3	40	250	9.5
					Ti-2.0I	900	3300	2.3	40	480	9.4
K6§§	5.00	2.01		1300 K/173 ks	Ti-1.2U	1300	3400	2.2	30	90	10
	(0.036)	(0.0062)		$x_{La_2O_3} = 0.19$	Ti-2.0I	900	3400	2.0	40	190	9.2
					Ti-2.0I	900	3400	2.1	40	190	9.2
Q10¶¶	5.12		25.0	1200 K/86 ks	Ti-1.2U	1300	60	18	30	840	8.8
	(0.037)		(0.10)	TiO_2 : 0.19 g (0.0024 mol)	Ti-2.0I	900	150	5.7	40	610	8.8
				$x_{LaOCl} = 0.047$							
M2††	2.97	1.00	2.02	1200 K/173 ks	Ti-1.2U	1300	100	6.3	30	50	10
	(0.021)	(0.0031)	(0.0082)	$x_{LaOCl} = 0.65$	Ti-2.0I	900	70	7.7	40	50	10
					Ti-2B	200	90	8.3	20	20	13
M3††	2.93	1.50	2.01	1200 K/173 ks	Ti-1.2U	1300	160	5.4	30	480	8.9
	(0.021)	(0.0046)	(0.0082)	$x_{LaOCl} = 0.79$	Ti-2.0I	900	120	6.2	40	40	11
					Ti-2.0I	900	70	9.6	40	80	9.8
N1¶	3.05		2.01	1200 K/173 ks	Ti-1.2U	1300	90	13	30	390	9.0
	(0.022)		(0.0082)	$x_{LaOCl} = 0$	Ti-2.0I	900	80	13	40	390	9.0
					Ti-2B	200	80	11	20	50	10
K1§§	2.92		2.00	1300 K/173 ks	Ti-1.2U	1300	40	44	30	140	9.4
	(0.021)		(0.0082)	$x_{LaOCl} = 0$	Ti-2.0I	900	40	34	40	310	9.0
					Ti-2B	200	40	28	20	60	9.7
K2§§	2.93	0.50	2.01	1300 K/173 ks	Ti-1.2U	1300	60	34	30	110	9.8
	(0.021)	(0.0015)	(0.0082)	$x_{LaOCl} = 0.40$	Ti-2.0I	900	30	51	40	100	9.8
					Ti-2B	200	50	25	20	220	9.1

Table III. (Continued)

Exp. No.	Initial Quantity of Samples in the Titanium Crucible*, m/g (n/mol)			Experimental Condition**	Type of Ti Samples†	Oxygen Concentration, C_O (mass ppm O)		Analyzed Error, ε_O (pct)	Nitrogen Concentration, C_N (mass ppm N)		Analyzed Error, ε_N (pct)
	La	La_2O_3	$LaCl_3$			Initial, $C_{O,\ initial}$	After Exp., $C_{O,\ after}$		Initial, $C_{N,\ initial}$	After exp., $C_{N,\ after}$	
K4§§	2.93 (0.021)	1.51 (0.0046)	2.01 (0.0082)	1300 K/173 ks x_{LaOCl}= 0.79	Ti-1.2U	1300	50	33	30	700	8.9
					Ti-2.0I	900	40	43	40	80	9.9
					Ti-2B	200	50	28	20	350	9.0

*Several Ti wires and shots (0.5 ~ 3.6 g in total) were inserted into the feedstock as monitors for p_{O_2}.

**x_i : the mole fraction of i, n_i : the number of mole of i $x_{La_2O_3}=\frac{n_{La_2O_3}}{n_{La_2O_3}+n_{La}}$, $x_{LaOCl}=\frac{n_{LaOCl}}{n_{LaOCl}+n_{LaCl_3}}$. It is assumed that the reactions as below were completely occurred.
1/3 La_2O_3 + 1/3 $LaCl_3$ → LaOCl; 1/2 TiO_2 + 2/3 La + 1/3 $LaCl_3$ → 1/2 Ti + LaOCl.

†Ti samples: 0.1 g, 1.2U: wire with 1.2 mm in diameter, 2.0I: wire with 2 mm in diameter, 2B: cube with 2 mm in length of side.

††Analytical conditions: Standard sample: Steel pin: 366 ± 6 ppm O, 23 ± 2 ppm N in 1.0 g, part number: 501-874; Graphite crucible: 0.1 ± 0.1 μg O, 0.0 ± 0.0 μg N in 1.0 g, part number: 782-720; Ni flux: 2.5 ± 0.5 μg O, 0.0 ± 0.0 μg N in 1.0 g, part number: 502-344.

¶Analytical conditions: Standard sample: Steel pin: 366 ± 6 ppm O, 23 ± 2 ppm N in 1.0 g, part number: 501-874; Graphite crucible: 0.1 ± 0.1 μg O, 0.0 ± 0.0 μg N in 1.0 g, part number: 782-720; Ni flux: 2.3 ± 0.8 μg O, 0.0 ± 0.0 μg N in 1.0 g, part number: 502-344.

¶¶Analytical conditions: Standard sample: Steel pin: 366 ± 6 ppm O, 23 ± 2 ppm N in 1.0 g, part number: 501-874; Graphite crucible: 0.1 ± 0.1 μg O, 0.0 ± 0.0 μg N in 1.0 g, part number: 782-720; Ni flux: 2.5 ± 0.8 μg O, 0.0 ± 0.0 μg N in 1.0 g, part number: 502-344.

§Analytical conditions: Standard sample: Steel pin: 365 ± 7 ppm O, 22 ± 2 ppm N in 1.0 g, part number: 501-646; Graphite crucible: 0.1 ± 0.1 μg O, 0.0 ± 0.0 μg N in 1.0 g, part number: 782-720; Ni flux: 3 ± 2 μg O, 0.0 ± 0.0 μg N in 1.0 g, part number: 502-344.

§§Analytical conditions: Standard sample: Steel pin: 366 ± 6 ppm O, 23 ± 2 ppm N in 1.0 g, part number: 501-874; Graphite crucible: 0.1 ± 0.1 μg O, 0.0 ± 0.0 μg N in 1.0 g, part number: 782-720; Ni flux: 2.5 ± 0.8 μg O, 0.0 ± 0.0 μg N in 1.0 g, part number: 502-344.

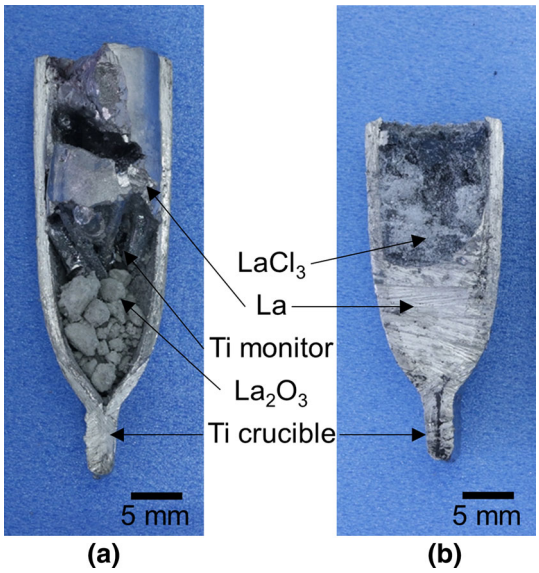


Fig. 7—The photographic images of the cross-sectioned Ti crucibles used in the deoxidation experiment; (*a*) in La/La_2O_3 equilibrium and (*b*) in La/LaOCl/$LaCl_3$ equilibrium.

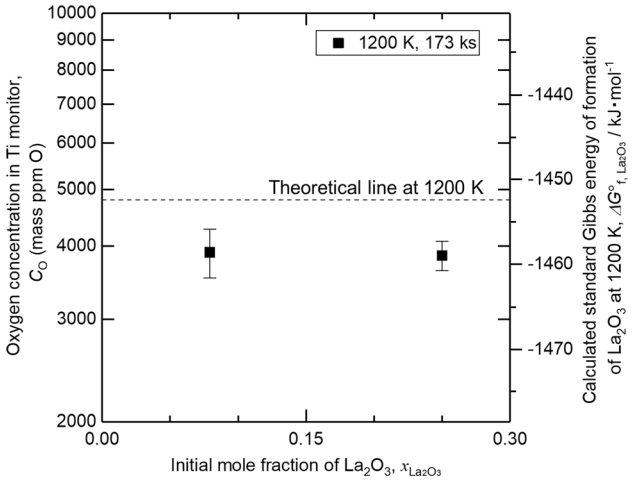


Fig. 8—The oxygen concentration in obtained Ti after lanthanothermic deoxidation experiments against the initial mole fraction of La_2O_3. The theoretical oxygen content line is also shown.

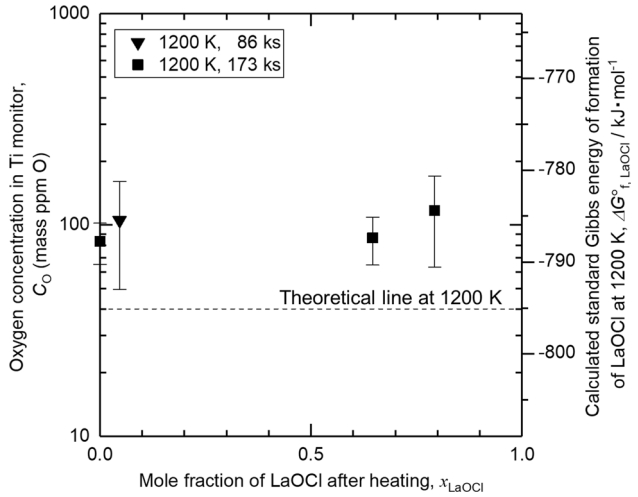


Fig. 9—The oxygen content in obtained Ti after lanthanothermic deoxidation experiments against the mole fraction of LaOCl after heating. The theoretical oxygen content line is also shown.

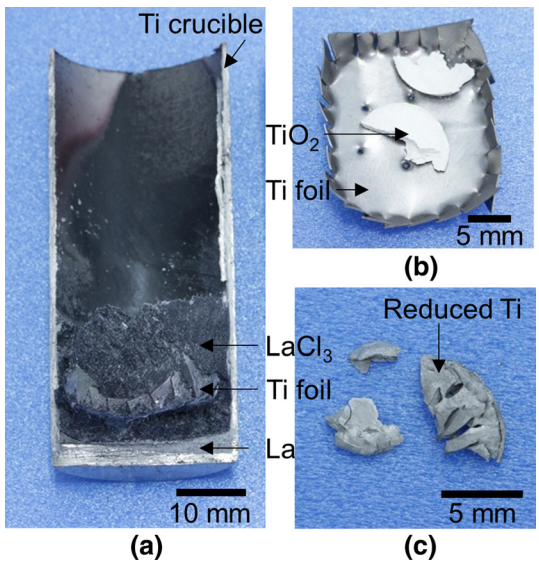


Fig. 10—The photographic images of the samples; (*a*) Ti crucible used in the reduction experiment, (*b*) TiO_2 pellets and Ti foil before experiment, and (*c*) the pellets after experiment.

with varying $x_{La_2O_3}$. Therefore, it is deduced that during the absorption of O into Ti, the system did not run out of O (*i.e.*, $a_{La_2O_3} = 1$). This result proves that it is impossible to deoxidize Ti to an O concentration of approximately 500 mass ppm O with the La/La_2O_3 equilibrium at 1200 K (927 °C).

Figure 9 shows the relationship between the deoxidation results of the Ti pieces and the molar fraction of LaOCl, with x_{LaOCl} at La/LaOCl/$LaCl_3$ equilibrium at 1200 K (927 °C); x_{LaOCl} is defined as $x_{LaOCl} = \frac{n_{LaOCl}}{n_{LaOCl} + n_{LaCl_3}}$, where n_{LaOCl} and n_{LaCl_3} are calculated from the initial amounts of La_2O_3 and $LaCl_3$, respectively, assuming that the reaction 1/3 La_2O_3 + 1/3 $LaCl_3$ = LaOCl occurred completely in the system. The oxygen concentration in Ti decreased to approximately 120 ± 50 mass ppm O, as expected from the thermodynamic calculation (Table III and Figure 9). From Figure 9, it is deduced that the LaOCl in the system was saturated because no increase or decrease in the oxygen concentration in Ti was observed with varying x_{LaOCl}. This result demonstrates that it is possible to decrease the oxygen concentration in Ti to less than 200 mass ppm O by using La as the deoxidant in $LaCl_3$ molten salt.

Furthermore, when the deoxidation reaction temperature was 1300 K (1027 °C), the oxygen concentration in the produced Ti was 3300 ± 200 mass ppm O and 50 ± 30 mass ppm O at La/La_2O_3 and La/LaOCl/$LaCl_3$ equilibria, respectively. Although the value of $\Delta G^\circ_{f,LaOCl}$

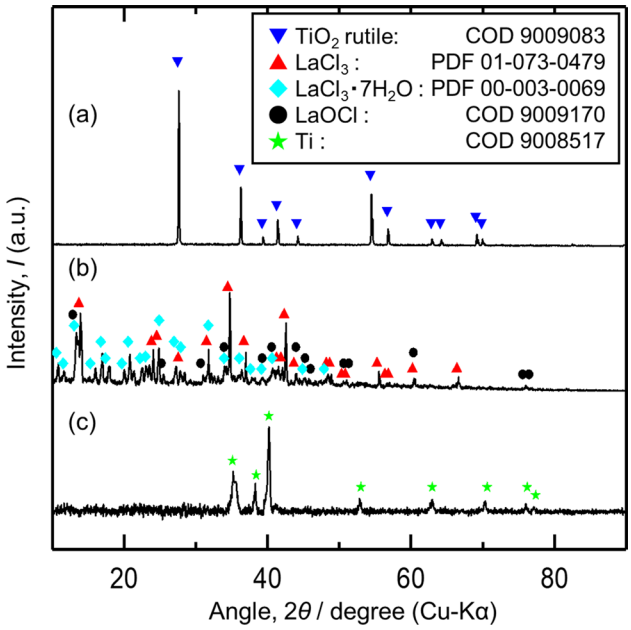


Fig. 11—XRD profiles of the experimental samples; (*a*) TiO_2 reagent sample heated at 1300 K for 86 ks, (*b*) salt removed from the heated Ti crucible, and (*c*) pellet reduced in $LaCl_3$ at 1200 K for 86 ks using La as the reductant.

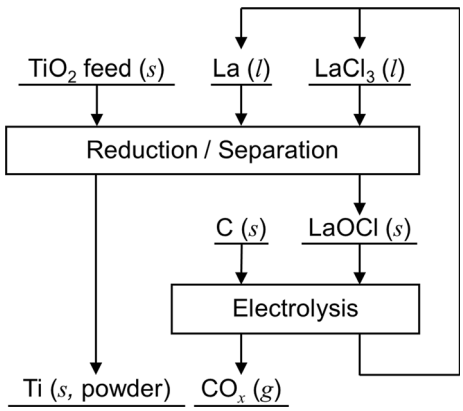


Fig. 12—The proposed Ti smelting process flow using the lanthanothermic reduction method of TiO_2.

at 1300 K (1027 °C) is not available, the decrease in the oxygen concentration of Ti in $LaCl_3$ molten salt is attributed to the formation of LaOCl.

The above results show that it is possible to deoxidize Ti to low-oxygen-concentration Ti by using La as the reductant and exploiting the LaOCl formation reaction.

B. *Reduction Experiment of TiO_2*

The deoxidation experiment of Ti described in Section IV–A showed that low-oxygen-concentration Ti can be produced at La/LaOCl/$LaCl_3$ equilibrium, and it is impossible to deoxidize Ti at La/La_2O_3 equilibrium. Based on these results, the reduction experiment of TiO_2 was conducted using La as the reductant in $LaCl_3$ molten salt.

Figure 10(a) shows the cross-section of the Ti crucible after heating. As shown at the bottom of Figure 10(a), La remained in the crucible after the experiment. Figures 10(b) and (c) show the TiO_2 pellets before and after the reduction reaction, respectively. The TiO_2 pellets, which had a white color before the experiment, were reduced to silvery pellets with a metallic luster.

Figure 11 shows the result of the XRD analysis. As shown in the figure, LaOCl, $LaCl_3$, and $LaCl_3{\cdot}7H_2O$ were detected in the salt, and Ti was detected in the pellets. Ti is considered to have been generated by the reduction reaction of TiO_2 using La as the reductant with the formation of LaOCl.

To evaluate the oxygen concentration of the obtained Ti pellets, the oxygen analysis was performed on the Ti pieces instead of the Ti pellets themselves. The oxygen content in these Ti pieces was found to be 100 ± 50 mass ppm O, which is consistent with the result of the deoxidation experiment (Section IV–A). Therefore, the oxygen concentration of the pellets, which are powder-sintered bodies and have specific surface areas relatively larger than that of the wire, is expected to be as low as 200 mass ppm O or less.

V. CONCLUSION

In this study, the authors developed a lanthanothermic reduction method of TiO_2. First, La was used to deoxidize metallic Ti, and it was demonstrated that the oxygen concentration of Ti decreased to less than 200 mass ppm O at 1200 K (927 °C) in $LaCl_3$ molten salt. Subsequently, metallic Ti was produced through the reduction of TiO_2 using La as the reductant in a $LaCl_3$ molten salt. The oxygen concentration of the Ti pieces placed in a crucible together with TiO_2 pellets was 100 ± 50 mass ppm O. The results indicate that it is possible to directly produce Ti with an oxygen concentration lower than 200 mass ppm O from TiO_2 by using La as the reductant in a $LaCl_3$ molten salt at 1200 K (927 °C). It was shown experimentally for the first time that the lanthanothermic reduction method is effective for the direct reduction of TiO_2 to low-oxygen-concentration Ti.

If in the future, the by-product of the lanthanothermic reduction, *i.e.*, LaOCl, can be recycled effectively, a process with zero-consumption of La could be designed, as shown in Figure 12. The results of this work are expected to lead to the development of an industrial production process of extremely low-oxygen-concentration Ti from Ti ore without the need for a chlorination process.

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